# ADS T1 ASSIGNMENT REPORT

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## COMPARATIVE ANALYSIS

### DATA INTEGRATION

```
data['Type'] = 'Apple'
data1['Type'] = 'Google'

data = data.rename(columns={'App_Name': 'App Name'})

apple_columns = ['App Name', 'Size_Bytes', 'Price', 'Average_User_Rating',
'Required_IOS_Version', 'Primary_Genre',
'ContentRating','Free','Released_Year','Updated_Year','Type']
google_columns = ['App Name', 'Size (MB)', 'Price', 'Rating', 'Minimum Android', 'Category',
'Content Rating','Free','Released_Year','Updated_Year','Type']

data2 = pd.merge(data[apple_columns], data1[google_columns], on=['App Name'], how='inner')

print(data2.head())
```

```
        App Name  Size_Bytes  Price_x  Average_User_Rating  \
0  A+ Paper Guide   20.974609      0.0                    0
1    A2 Directory   65.580078      0.0                    0
2        A4 News   13.184570      0.0                    0
3     B & O Books   77.805664      0.0                    0
4       B&B Cafe   45.814453      0.0                    5

   Required_IOS_Version Primary_Genre ContentRating  Free_x  Released_Year_x  \
0                   8.0     Education       Toddler       1             2017
1                  11.0          Book       Toddler       1             2020
2                  13.0          News       Toddler       1             2019
3                   8.0          Book       Toddler       1             2020
4                  12.0          Book       Toddler       1             2021

  Updated_Year_x  ... Size (MB)  Price_y  Rating  Minimum Android  \
0           2018  ...       9.4      0.0     0.0              4.4
```

```
1          2021  ...      42.0      0.0     5.0             5.1
2          2020  ...       7.4      0.0     0.0             5.0
3          2020  ...      20.0      0.0     0.0             5.0
4          2021  ...      15.0      0.0     0.0             4.1

          Category Content Rating Free_y  Released_Year_y Updated_Year_y  \
0          Education       Everyone      1            2017          2018
1             Social          Teen       1            2020          2020
2   News & Magazines       Everyone      1            2019          2020
3  Books & Reference       Everyone      1            2020          2020
4       Food & Drink       Everyone      1            2021          2021

   Type_y
0  Google
1  Google
2  Google
3  Google
4  Google

[5 rows x 21 columns]


data2.dtypes

App Name                object
Size_Bytes             float64
Price_x                float64
Average_User_Rating      int64
Required_IOS_Version   float64
Primary_Genre           object
ContentRating           object
Free_x                  int64
Released_Year_x         int64
Updated_Year_x          object
Type_x                  object
Size (MB)              float64
Price_y                float64
Rating                 float64
Minimum Android         object
Category                object
Content Rating          object
Free_y                  int64
Released_Year_y         object
Updated_Year_y          object
```

Type_y object
dtype: object

```
data2 = data2.sort_values(by='App Name')
data2 = data2.drop_duplicates(subset=['App Name'])
```

```
data2.head()
```

| | App Name | Size_Bytes | Price_x | Average_User_Rating | Required_IOS_Version | Primary_Genre | ContentRating | Free_x | Released_Year_x | Updated_Year_x | ... | Size (MB) | Price_y | Rating | Minimum Android | Category | Content Rating | Free_y | Released_Year_y | Updated_Year_y | Type_y |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 327808 | 14F CataloniaÂ´s Elections | 21.886719 | 0.0 | 0 | 13.0 | News | Toddler | 1 | 2021 | 2021 | ... | 20.0 | 0.0 | 3.1 | 7.0 | News & Magazines | Everyone | 1 | 2021 | 2021 | Google |
| 77004 | 3D Meiotic Division | 193.672852 | 0.0 | 0 | 9.0 | Education | Toddler | 1 | 2019 | 2019 | ... | 74.0 | 0.0 | 4.1 | 4.4 | Education | Everyone | 1 | 2019 | 2020 | Google |
| 77005 | 3D Mitotic Division | 154.509766 | 0.0 | 3 | 7.0 | Education | Toddler | 1 | 2017 | 2019 | ... | 60.0 | 0.0 | 4.5 | 4.4 | Education | Everyone | 1 | 2017 | 2020 | Google |
| 44254 | A Academy | 19.334961 | 0.0 | 0 | 8.0 | Education | Toddler | 1 | 2017 | 2017 | ... | 2.8 | 0.0 | 0.0 | 4.1 | Education | Everyone | 1 | 2016 | 2018 | Google |
| 113759 | A Ambassador Limousine | 61.852539 | 0.0 | 0 | 10.3 | Business | Toddler | 1 | 2017 | 2021 | ... | 4.1 | 0.0 | 0.0 | 5.0 | Travel & Local | Everyone | 1 | 2017 | 2020 | Google |

5 rows × 21 columns

```
has_duplicate_rows = data2.duplicated().any()
print(has_duplicate_rows)
```

False

```
data2.shape
```

(363614, 21)

```
data2
```

| | App Name | Size_Bytes | Price_x | Average_User_Rating | Required_IOS_Version | Primary_Genre | ContentRating | Free_x | Released_Year_x | Updated_Year_x | ... | Size (MB) | Price_y | Rating | Mi An |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 327808 | 14F CataloniaÂ´s Elections | 21.886719 | 0.0 | 0 | 13.0 | News | Toddler | 1 | 2021 | 2021 | ... | 20.0 | 0.0 | 3.1 | |
| 77004 | 3D Meiotic Division | 193.672852 | 0.0 | 0 | 9.0 | Education | Toddler | 1 | 2019 | 2019 | ... | 74.0 | 0.0 | 4.1 | |
| 77005 | 3D Mitotic Division | 154.509766 | 0.0 | 3 | 7.0 | Education | Toddler | 1 | 2017 | 2019 | ... | 60.0 | 0.0 | 4.5 | |
| 44254 | A Academy | 19.334961 | 0.0 | 0 | 8.0 | Education | Toddler | 1 | 2017 | 2017 | ... | 2.8 | 0.0 | 0.0 | |
| 113759 | A Ambassador Limousine | 61.852539 | 0.0 | 0 | 10.3 | Business | Toddler | 1 | 2017 | 2021 | ... | 4.1 | 0.0 | 0.0 | |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | |
| 23134 | myCirqa | 6.284180 | 0.0 | 0 | 9.0 | Book | Toddler | 1 | 2016 | 2019 | ... | 2.8 | 0.0 | 2.5 | |
| 1499 | mySAPL | 26.045898 | 0.0 | 0 | 11.0 | Reference | Toddler | 1 | 2015 | 2021 | ... | 4.4 | 0.0 | 3.9 | |
| 375180 | swaarm | 28.590820 | 0.0 | 0 | 9.0 | Lifestyle | Toddler | 1 | 2021 | 2021 | ... | 34.0 | 0.0 | 0.0 | |
| 279872 | Ä°stanbul Airport | 148.147461 | 0.0 | 4 | 11.0 | Travel | Toddler | 1 | 2019 | 2021 | ... | 59.0 | 0.0 | 3.8 | |
| 251649 | é□□ë□ æ□ á□□ | 69.065430 | 0.0 | 0 | 9.0 | Health & Fitness | Toddler | 1 | 2018 | 2019 | ... | 11.0 | 0.0 | 0.0 | |

363614 rows × 21 columns

```
data2['App Name'].value_counts()
```

```
14F CataloniaÂ´s Elections    1
Pearl River Coffee News      1
Pear reSET-OÂ®                1
Pear Training                1
Pear Street Bistro Pinole    1
                            ..
Frugal                       1
Fruchthandel Magazin         1
FruchtXpress                 1
Frubble                      1
é□□é□æ□ å□□                  1
Name: App Name, Length: 363614, dtype: int64
```

```
data2.columns
```

```
Index(['App Name', 'Size_Bytes', 'Price_x', 'Average_User_Rating',
       'Required_IOS_Version', 'Primary_Genre', 'ContentRating', 'Free_x',
       'Released_Year_x', 'Updated_Year_x', 'Type_x', 'Size (MB)', 'Price_y',
       'Rating', 'Minimum Android', 'Category', 'Content Rating', 'Free_y',
       'Released_Year_y', 'Updated_Year_y', 'Type_y'],
      dtype='object')
```

```
data2['Primary_Genre'].unique()
```

```
array(['News', 'Education', 'Business', 'Book', 'Medical', 'Productivity',
       'Shopping', 'Travel', 'Health & Fitness', 'Utilities', 'Reference',
       'Sports', 'Lifestyle', 'Finance', 'Food & Drink', 'Navigation',
       'Music', 'Games', 'Social Networking', 'Entertainment',
       'Photo & Video', 'Magazines & Newspapers', 'Weather', 'Stickers',
       'Graphics & Design', 'Developer Tools'], dtype=object)
```

```
data2['Category'].unique()
```

```
array(['News & Magazines', 'Education', 'Travel & Local',
       'Books & Reference', 'Business', 'Educational', 'Medical',
       'Maps & Navigation', 'Shopping', 'Health & Fitness',
       'Productivity', 'Entertainment', 'Auto & Vehicles', 'Beauty',
       'Lifestyle', 'Finance', 'Food & Drink', 'Communication', 'Events',
       'House & Home', 'Parenting', 'Music & Audio', 'Arcade', 'Puzzle',
       'Tools', 'Sports', 'Social', 'Simulation', 'Role Playing',
       'Photography', 'Casual', 'Libraries & Demo', 'Weather',
       'Video Players & Editors', 'Word', 'Dating', 'Action',
       'Personalization', 'Comics', 'Racing', 'Trivia', 'Card',
       'Strategy', 'Casino', 'Adventure', 'Art & Design', 'Board',
       'Music'], dtype=object)
```

**COMPARATIVE ANALYSIS**

```
! pip install fuzzywuzzy
```

```
Requirement already satisfied: fuzzywuzzy in /usr/local/lib/python3.10/dist-packages (0.18.0)
```

```
from fuzzywuzzy import fuzz

threshold = 30

def compare_genre_and_category(row):

    google_category = row['Category']
    apple_genre = row['Primary_Genre']

    similarity_ratio = fuzz.token_set_ratio(google_category.lower(), apple_genre.lower())
    return similarity_ratio >= threshold

data2['Genre_vs_Category'] = data2.apply(compare_genre_and_category, axis=1)

print(data2[['App Name', 'Category', 'Primary_Genre', 'Genre_vs_Category']])
```

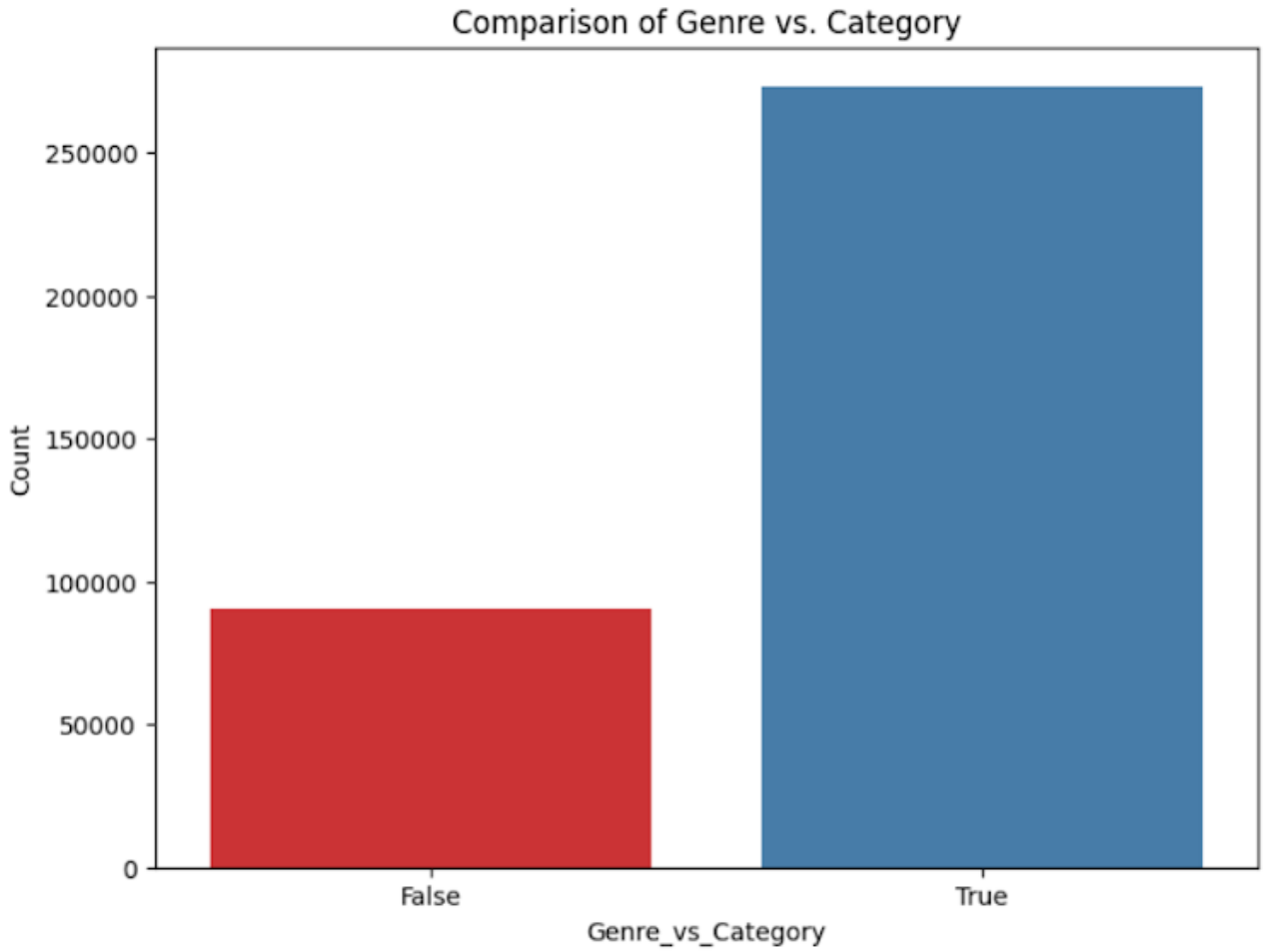
```
                   App Name           Category     Primary_Genre  \
327808  14F CataloniaÂ´s Elections   News & Magazines            News
77004          3D Meiotic Division          Education         Education
77005          3D Mitotic Division          Education         Education
44254                  A Academy          Education         Education
113759      A Ambassador Limousine     Travel & Local          Business
...                        ...                ...               ...
23134                   myCirqa   Libraries & Demo              Book
1499                    mySAPL  Books & Reference         Reference
375180                   swaarm          Lifestyle         Lifestyle
279872          Ä°stanbul Airport     Travel & Local            Travel
251649               é□□é□æ□ å□□             Tools  Health & Fitness

       Genre_vs_Category
327808              True
77004               True
77005               True
44254               True
113759             False
...                  ...
23134              False
1499                True
```

375180 True
279872 True
251649 False

[363614 rows x 4 columns]

```
plt.figure(figsize=(8, 6))
sns.countplot(data=data2, x='Genre_vs_Category', palette='Set1')
plt.title('Comparison of Genre vs. Category')
plt.xlabel('Genre_vs_Category')
plt.ylabel('Count')
plt.show()
```



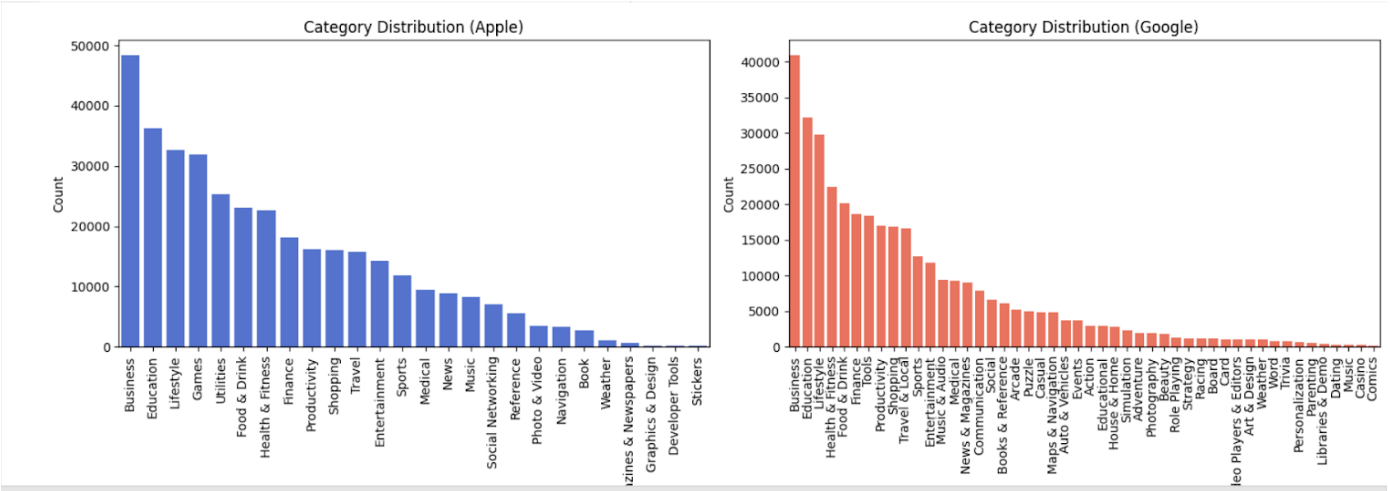
```
category_counts_apple = data2['Primary_Genre'][data2['Type_x'] == 'Apple'].value_counts()
category_counts_google = data2['Category'][data2['Type_y'] == 'Google'].value_counts()


apple_color = 'royalblue'
google_color = 'tomato'


plt.figure(figsize=(15, 6))
plt.subplot(1, 2, 1)
sns.barplot(x=category_counts_apple.index, y=category_counts_apple.values,
color=apple_color)
plt.title('Category Distribution (Apple)')
plt.xlabel('Primary Genre')
plt.ylabel('Count')
plt.xticks(rotation=90)

plt.subplot(1, 2, 2)
sns.barplot(x=category_counts_google.index, y=category_counts_google.values,
color=google_color)
plt.title('Category Distribution (Google)')
plt.xlabel('Category')
plt.ylabel('Count')
plt.xticks(rotation=90)

plt.tight_layout()
plt.show()
```



```
threshold = 30

def compare_content_rating(row):

    google_category = row['Content Rating']
    apple_genre = row['ContentRating']

    similarity_ratio = fuzz.token_set_ratio(google_category.lower(), apple_genre.lower())

    return similarity_ratio >= threshold


data2['Diff_Content_Rating'] = data2.apply(compare_content_rating, axis=1)

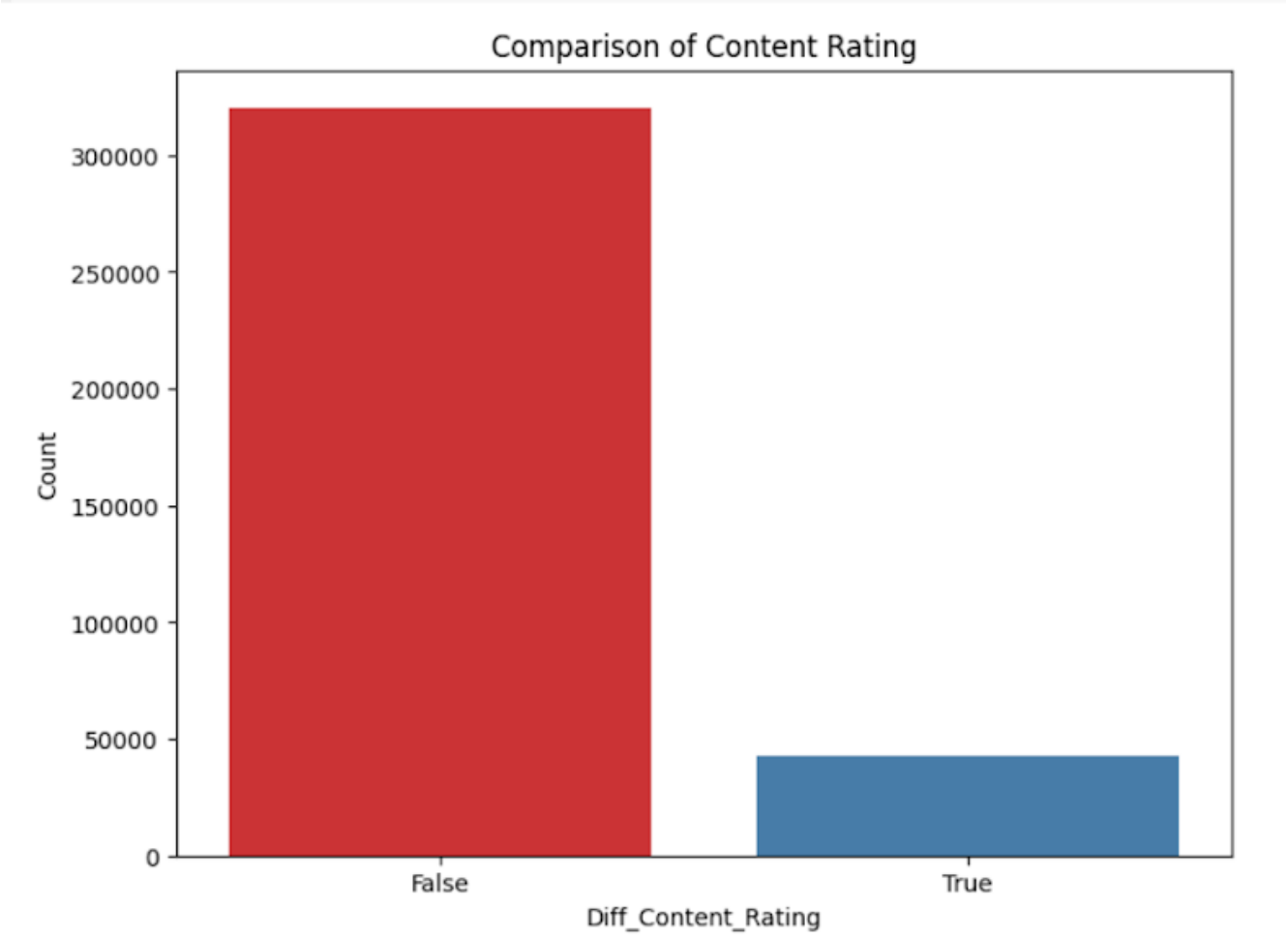
print(data2[['App Name', 'Content Rating', 'ContentRating', 'Diff_Content_Rating']])
```

```
                    App Name Content Rating ContentRating  \
327808  14F CataloniaÂ´s Elections       Everyone       Toddler
77004          3D Meiotic Division       Everyone       Toddler
77005          3D Mitotic Division       Everyone       Toddler
44254                    A Academy       Everyone       Toddler
113759      A Ambassador Limousine       Everyone       Toddler
...                          ...           ...          ...
23134                    myCirqa       Everyone       Toddler
1499                     mySAPL        Everyone       Toddler
375180                    swaarm       Everyone       Toddler
279872          Ä°stanbul Airport       Everyone       Toddler
251649               é□□é□æ□ å□□       Everyone       Toddler

       Diff_Content_Rating
327808               False
77004                False
77005                False
44254                False
113759               False
...                    ...
23134                False
1499                False
375180               False
279872               False
251649               False

[363614 rows x 4 columns]
```

```
plt.figure(figsize=(8, 6))
sns.countplot(data=data2, x='Diff_Content_Rating', palette='Set1')
plt.title('Comparison of Content Rating')
plt.xlabel('Diff_Content_Rating')
plt.ylabel('Count')
plt.show()
```



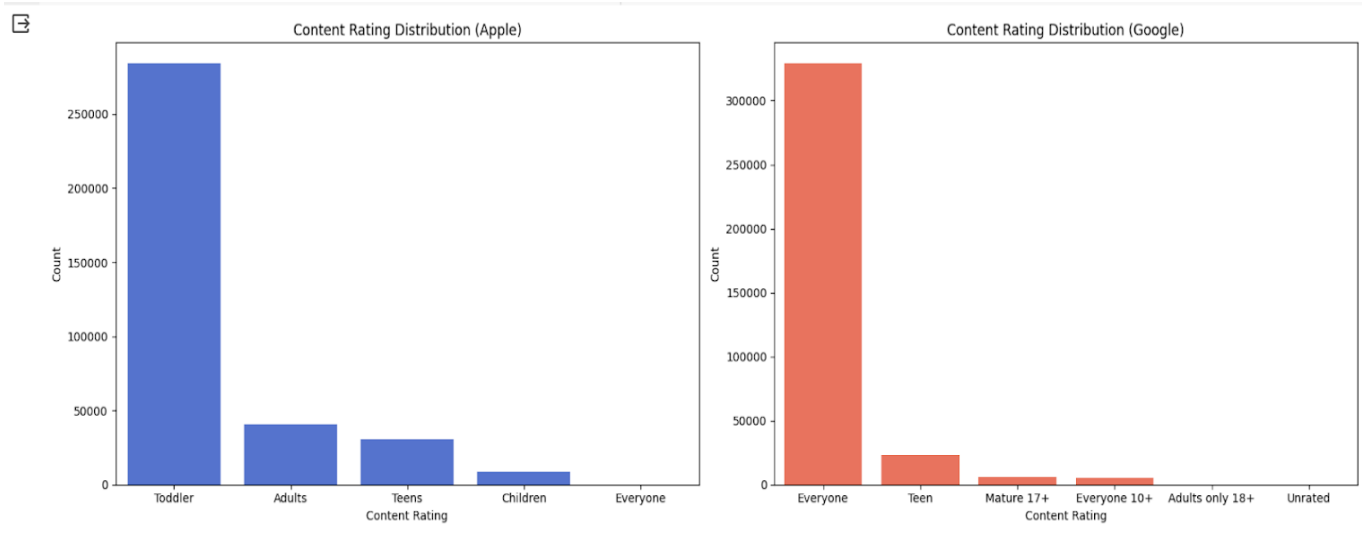
```
content_rating_counts_apple = data2['ContentRating'][data2['Type_x'] == 'Apple'].value_counts()
content_rating_counts_google = data2['Content Rating'][data2['Type_y'] ==
'Google'].value_counts()

apple_color = 'royalblue'
google_color = 'tomato'

plt.figure(figsize=(17, 6))
plt.subplot(1, 2, 1)
sns.barplot(x=content_rating_counts_apple.index, y=content_rating_counts_apple.values,
color=apple_color)
plt.title('Content Rating Distribution (Apple)')
plt.xlabel('Content Rating')
plt.ylabel('Count')

plt.subplot(1, 2, 2)
sns.barplot(x=content_rating_counts_google.index, y=content_rating_counts_google.values,
color=google_color)
plt.title('Content Rating Distribution (Google)')
plt.xlabel('Content Rating')
plt.ylabel('Count')

plt.tight_layout()
plt.show()
```



```
print(data2['Released_Year_x'].dtype)
print(data2['Released_Year_y'].dtype)
```

```
int64
object
```

```
data2['Released_Year_y'] = data2['Released_Year_y'].astype('int64')

data2['Earlier_Release_Platform'] = np.where(
    data2['Released_Year_x'] < data2['Released_Year_y'], 'Apple',
    np.where(data2['Released_Year_x'] > data2['Released_Year_y'], 'Google', 'Same')
)

print(data2[['App Name', 'Released_Year_x', 'Released_Year_y', 'Earlier_Release_Platform']])
```

```
                    App Name  Released_Year_x  Released_Year_y  \
327808  14F CataloniaÂ´s Elections             2021             2021
77004          3D Meiotic Division             2019             2019
77005          3D Mitotic Division             2017             2017
44254                    A Academy             2017             2016
113759       A Ambassador Limousine             2017             2017
...                             ...              ...              ...
23134                     myCirqa             2016             2017
1499                      mySAPL             2015             2015
375180                     swaarm             2021             2021
279872          Ä°stanbul Airport             2019             2019
251649              é□□é□æ□ å□□             2018             2019

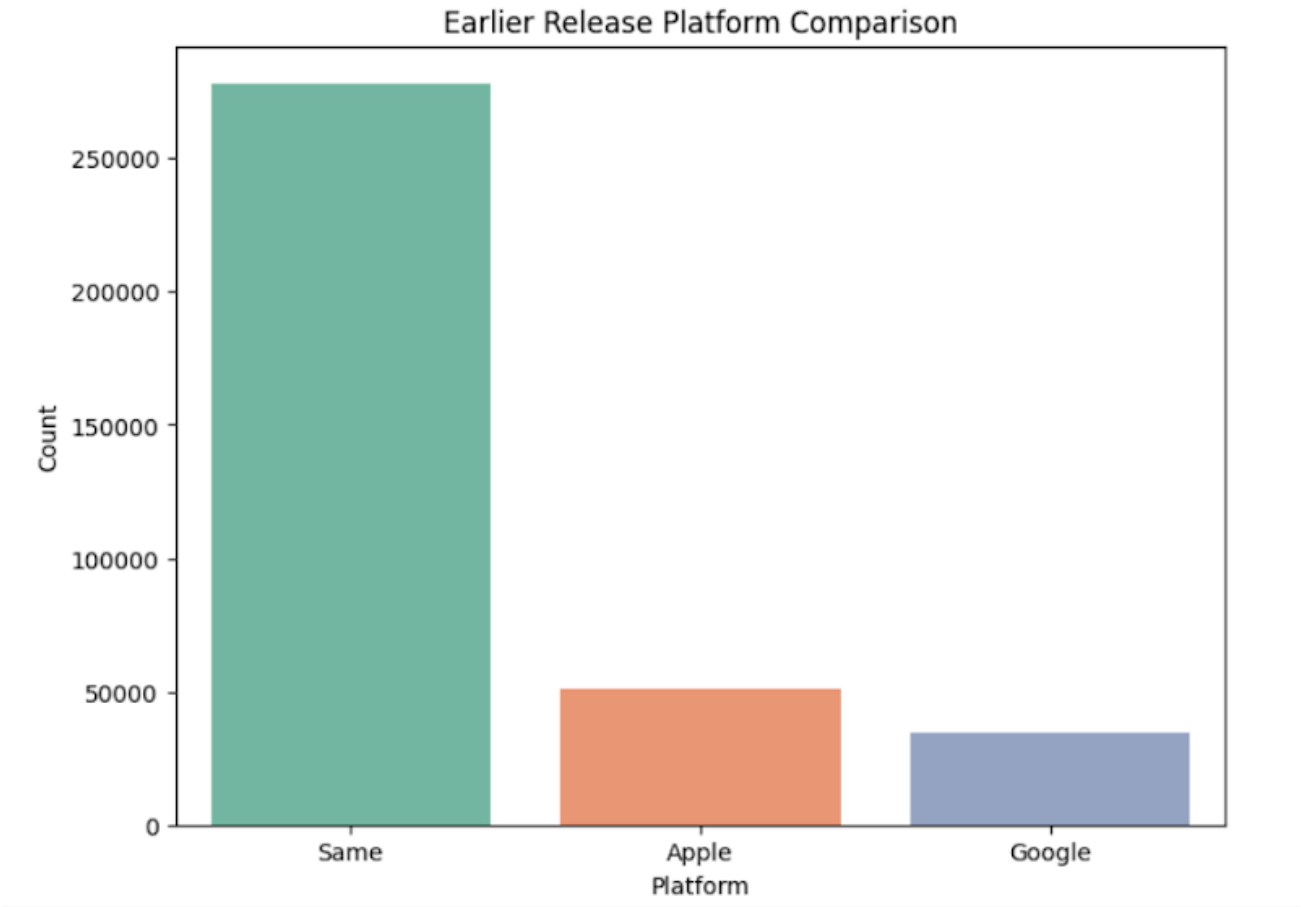
       Earlier_Release_Platform
327808                     Same
77004                      Same
77005                      Same
44254                    Google
113759                     Same
...                         ...
23134                    Apple
1499                       Same
375180                     Same
279872                     Same
251649                   Apple

[363614 rows x 4 columns]
```

```
platform_released_counts = data2['Earlier_Release_Platform'].value_counts()
print(platform_released_counts)

plt.figure(figsize=(8, 6))
sns.barplot(x=platform_released_counts.index, y=platform_released_counts.values,
palette='Set2')
plt.title('Earlier Release Platform Comparison')
plt.xlabel('Platform')
plt.ylabel('Count')
plt.show()
```

```
Same      278181
Apple      50858
Google     34575
Name: Earlier_Release_Platform, dtype: int64
```



```
data2['Earlier_Updated_Platform'] = np.where(
    data2['Updated_Year_x'] < data2['Updated_Year_y'], 'Apple',
    np.where(data2['Updated_Year_x'] > data2['Updated_Year_y'], 'Google', 'Same')
)

print(data2[['App Name', 'Updated_Year_x', 'Updated_Year_y', 'Earlier_Updated_Platform']])

                      App Name Updated_Year_x Updated_Year_y  \
327808  14F CataloniaÂ´s Elections          2021          2021
77004          3D Meiotic Division          2019          2020
77005          3D Mitotic Division          2019          2020
44254                    A Academy          2017          2018
113759       A Ambassador Limousine          2021          2020
...                             ...           ...           ...
23134                     myCirqa          2019          2019
1499                      mySAPL          2021          2021
375180                     swaarm          2021          2021
279872          Ä°stanbul Airport          2021          2021
251649               é□□é□æ□ å□□          2019          2019

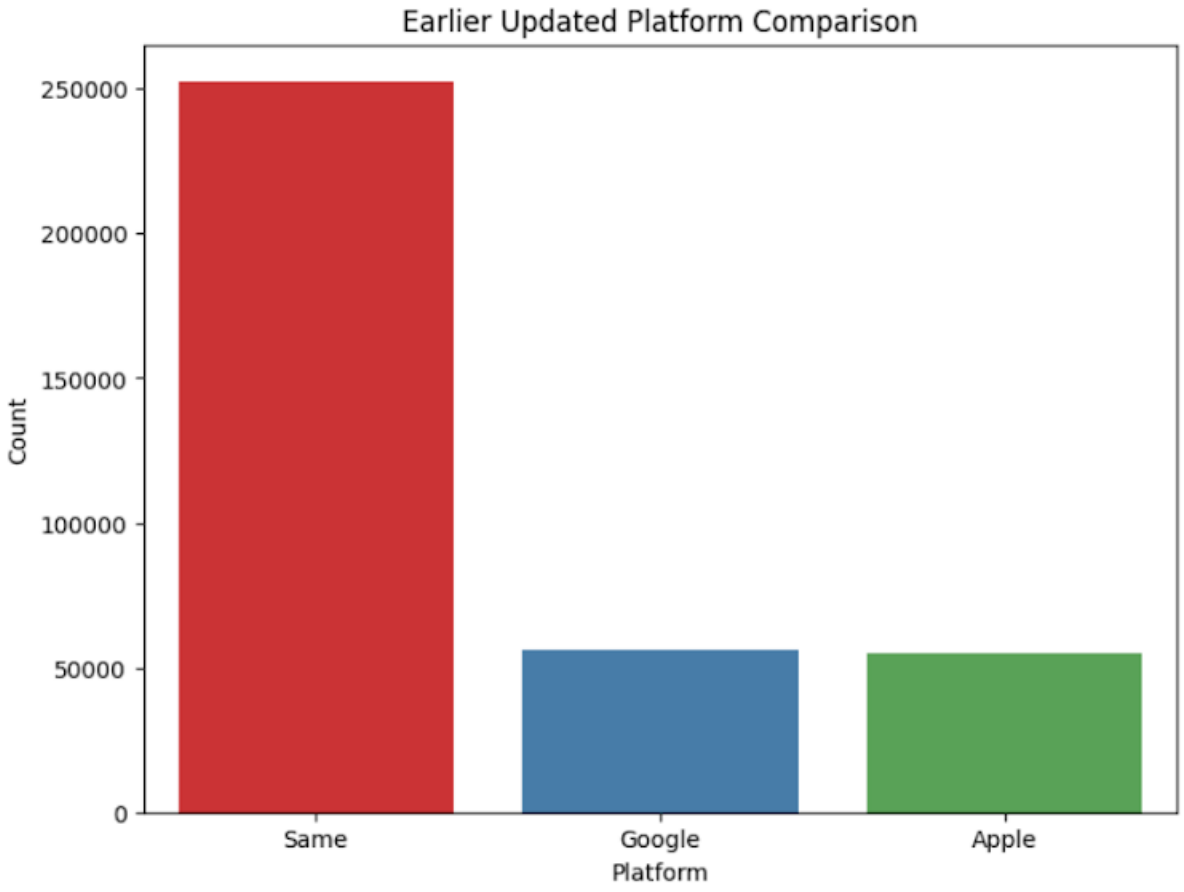
       Earlier_Updated_Platform
327808                     Same
77004                     Apple
77005                     Apple
44254                     Apple
113759                   Google
...                         ...
23134                      Same
1499                       Same
375180                     Same
279872                     Same
251649                     Same

[363614 rows x 4 columns]
```

```
platform_updated_counts = data2['Earlier_Updated_Platform'].value_counts()
print(platform_updated_counts)

plt.figure(figsize=(8, 6))
sns.barplot(x=platform_updated_counts.index, y=platform_updated_counts.values,
palette='Set1')
plt.title('Earlier Updated Platform Comparison')
plt.xlabel('Platform')
plt.ylabel('Count')
plt.show()
```

```
Same      252630
Google     56163
Apple      54821
Name: Earlier_Updated_Platform, dtype: int64
```



```
data2['Size_Comparison'] = np.where(
   data2['Size (MB)'] < data2['Size_Bytes'], 'Apple',
   np.where(data2['Size (MB)'] > data2['Size_Bytes'], 'Google', 'Same')
)

print(data2[['App Name', 'Size (MB)', 'Size_Bytes', 'Size_Comparison']])
```

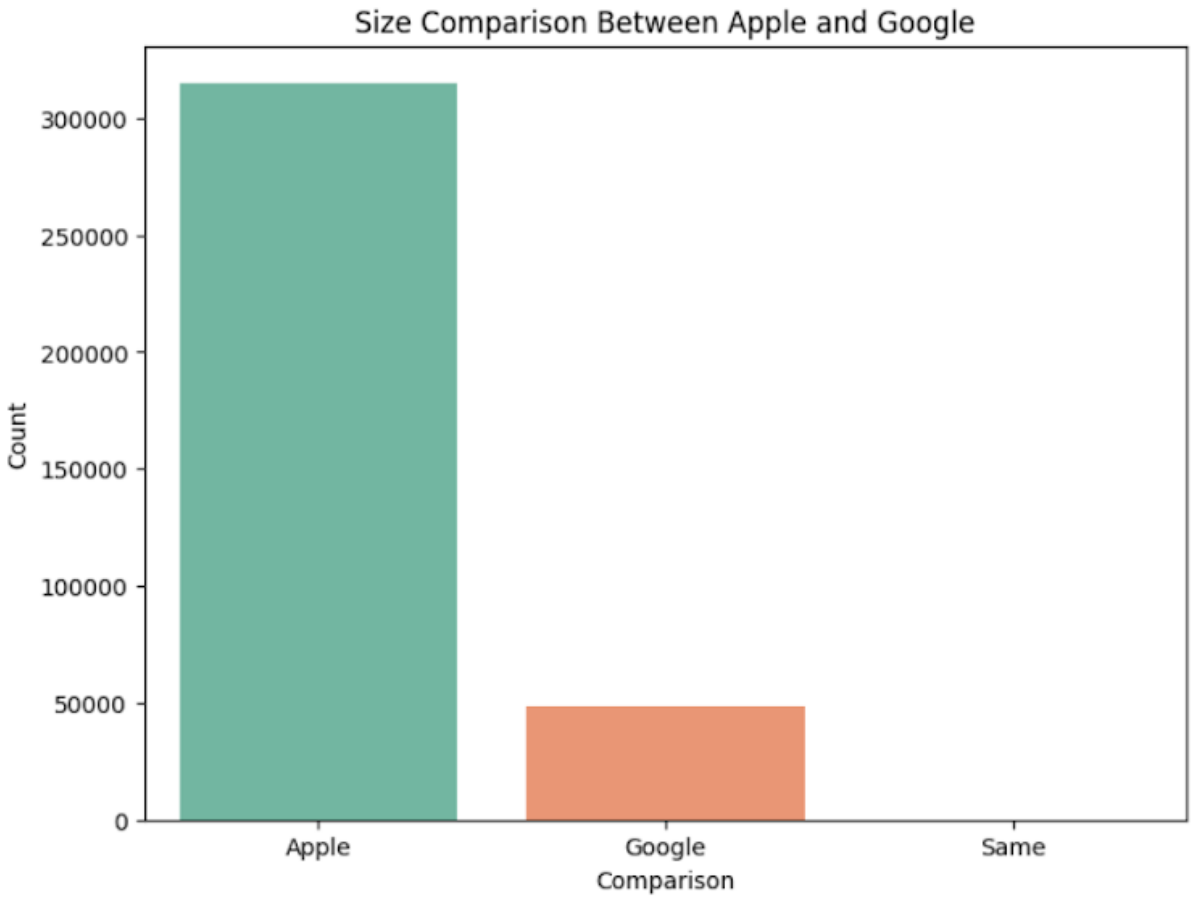
| | App Name | Size (MB) | Size_Bytes | Size_Comparison |
|---|---|---|---|---|
| 327808 | 14F CataloniaÂ´s Elections | 20.0 | 21.886719 | Apple |
| 77004 | 3D Meiotic Division | 74.0 | 193.672852 | Apple |
| 77005 | 3D Mitotic Division | 60.0 | 154.509766 | Apple |
| 44254 | A Academy | 2.8 | 19.334961 | Apple |
| 113759 | A Ambassador Limousine | 4.1 | 61.852539 | Apple |
| ... | ... | ... | ... | ... |
| 23134 | myCirqa | 2.8 | 6.284180 | Apple |
| 1499 | mySAPL | 4.4 | 26.045898 | Apple |
| 375180 | swaarm | 34.0 | 28.590820 | Google |
| 279872 | Ä°stanbul Airport | 59.0 | 148.147461 | Apple |
| 251649 | é□□é□æ□ å□□ | 11.0 | 69.065430 | Apple |

[363614 rows x 4 columns]

```
size_comparison_counts = data2['Size_Comparison'].value_counts()
print(size_comparison_counts)

plt.figure(figsize=(8, 6))
sns.barplot(x=size_comparison_counts.index, y=size_comparison_counts.values,
palette='Set2')
plt.title('Size Comparison Between Apple and Google')
plt.xlabel('Comparison')
plt.ylabel('Count')
plt.show()
```

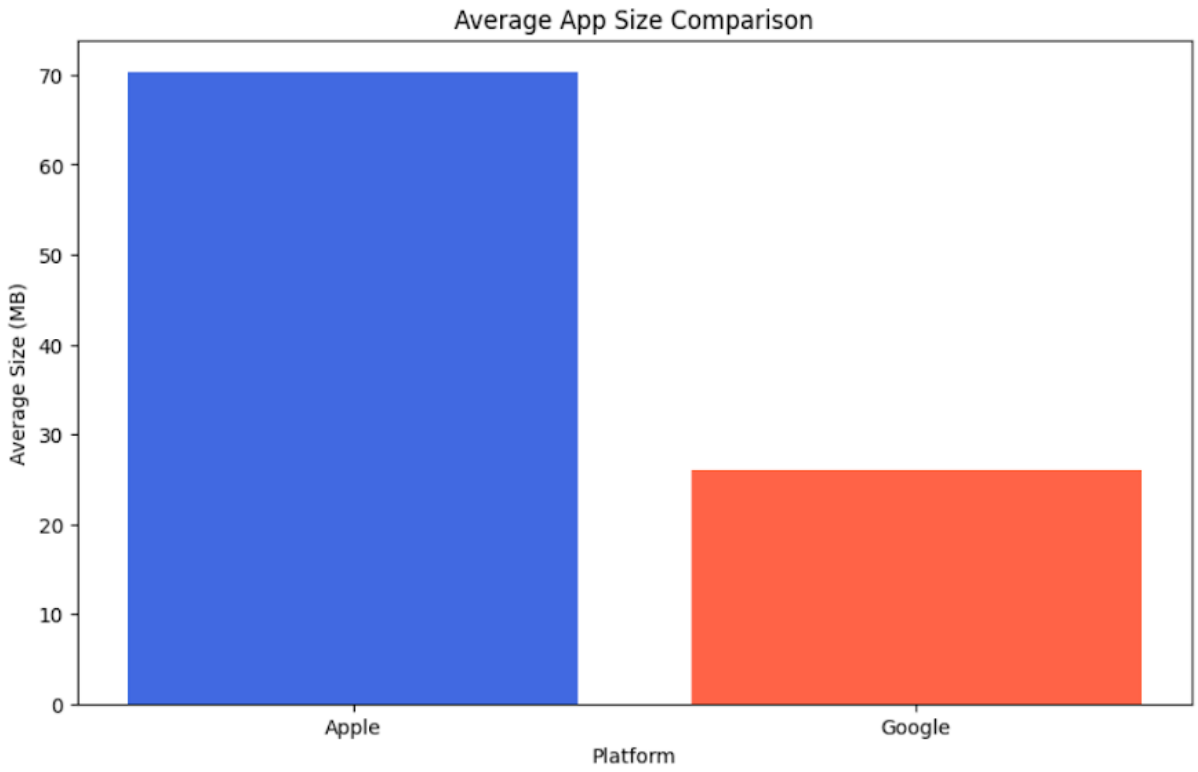
```
Apple     315121
Google     48490
Same           3
Name: Size_Comparison, dtype: int64
```



```
average_size_apple = data2['Size_Bytes'][data2['Type_x'] == 'Apple'].mean()
average_size_google = data2['Size (MB)'][data2['Type_y'] == 'Google'].mean()

apple_color = 'royalblue'
google_color = 'tomato'

plt.figure(figsize=(10, 6))
plt.bar(['Apple', 'Google'], [average_size_apple, average_size_google], color=[apple_color,
google_color])
plt.title('Average App Size Comparison')
plt.xlabel('Platform')
plt.ylabel('Average Size (MB)')
plt.show()
```



```
data2['Rating_Comparison'] = np.where(
   data2['Average_User_Rating'] > data2['Rating'], 'Apple',
   np.where(data2['Average_User_Rating'] < data2['Rating'], 'Google', 'Same')
)


print(data2[['App Name', 'Average_User_Rating', 'Rating', 'Rating_Comparison']])

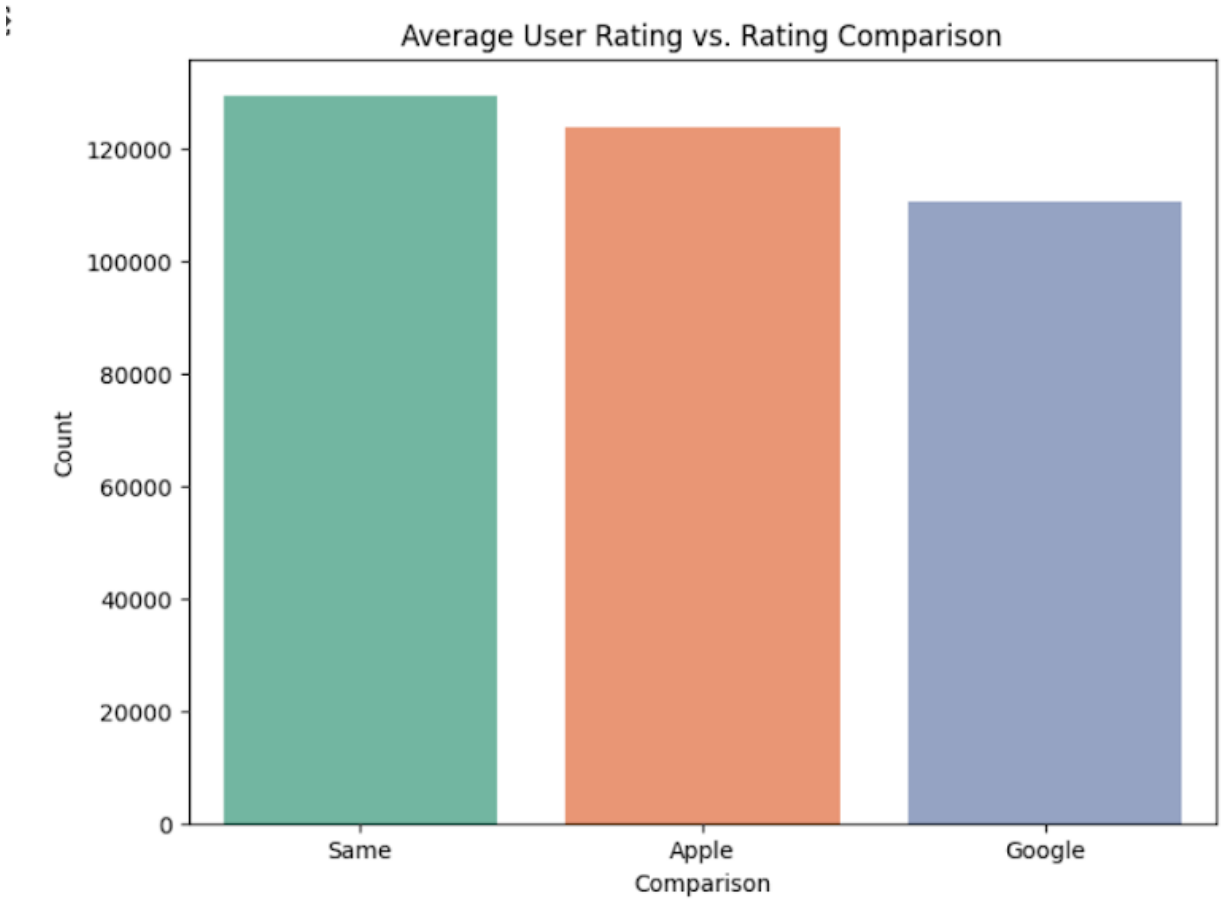

                    App Name  Average_User_Rating  Rating  \
327808  14F CataloniaÂ´s Elections                    0     3.1
77004          3D Meiotic Division                    0     4.1
77005          3D Mitotic Division                    3     4.5
44254                    A Academy                    0     0.0
113759       A Ambassador Limousine                     0     0.0
...                        ...                  ...     ...
23134                    myCirqa                    0     2.5
1499                     mySAPL                     0     3.9
375180                    swaarm                    0     0.0
279872          Ä°stanbul Airport                    4     3.8
251649              é□□é□æ□ å□□                      0     0.0

       Rating_Comparison
327808           Google
77004            Google
77005            Google
44254              Same
113759             Same
...                 ...
23134            Google
1499             Google
375180             Same
279872            Apple
251649             Same

[363614 rows x 4 columns]
```

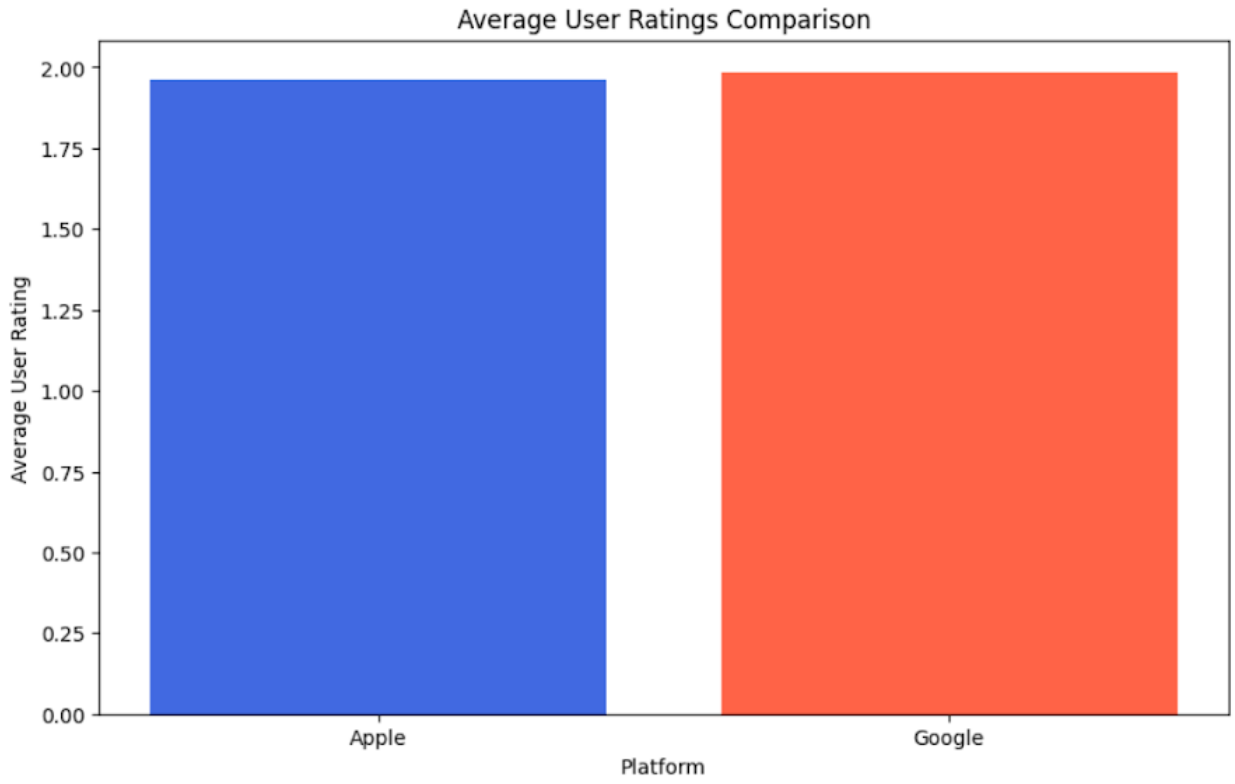
```
rating_comparison_counts = data2['Rating_Comparison'].value_counts()

plt.figure(figsize=(8, 6))
sns.barplot(x=rating_comparison_counts.index, y=rating_comparison_counts.values,
palette='Set2')
plt.title('Average User Rating vs. Rating Comparison')
plt.xlabel('Comparison')
plt.ylabel('Count')
plt.show()
```



```
apple_color = 'royalblue'
google_color = 'tomato'

plt.figure(figsize=(10, 6))
plt.bar(['Apple', 'Google'], [data2['Average_User_Rating'][data2['Type_x'] == 'Apple'].mean(),
data2['Rating'][data2['Type_y'] == 'Google'].mean()],  color=[apple_color, google_color])
plt.title('Average User Ratings Comparison')
plt.xlabel('Platform')
plt.ylabel('Average User Rating')
plt.show()
```



```
data2['Price_Comparison'] = np.where(
    data2['Price_y'] < data2['Price_x'], 'Apple',
    np.where(data2['Price_y'] > data2['Price_x'], 'Google', 'Same')
)

print(data2[['App Name', 'Price_x', 'Price_y', 'Price_Comparison']])
```

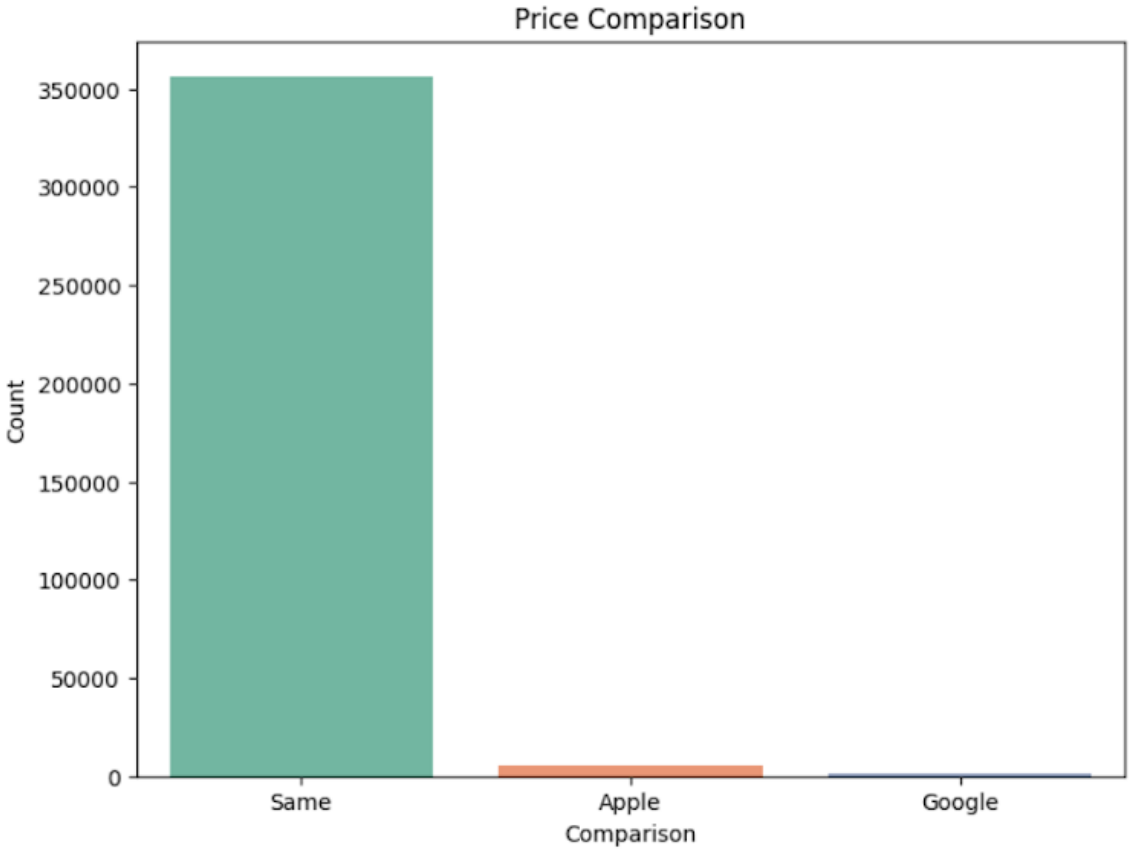
| | App Name | Price_x | Price_y | Price_Comparison |
|---|---|---|---|---|
| 327808 | 14F CataloniaÂ´s Elections | 0.0 | 0.0 | Same |
| 77004 | 3D Meiotic Division | 0.0 | 0.0 | Same |
| 77005 | 3D Mitotic Division | 0.0 | 0.0 | Same |
| 44254 | A Academy | 0.0 | 0.0 | Same |
| 113759 | A Ambassador Limousine | 0.0 | 0.0 | Same |
| ... | ... | ... | ... | ... |
| 23134 | myCirqa | 0.0 | 0.0 | Same |
| 1499 | mySAPL | 0.0 | 0.0 | Same |
| 375180 | swaarm | 0.0 | 0.0 | Same |
| 279872 | Ä°stanbul Airport | 0.0 | 0.0 | Same |
| 251649 | é□□é□æ□ å□□ | 0.0 | 0.0 | Same |

[363614 rows x 4 columns]

```
price_comparison_counts = data2['Price_Comparison'].value_counts()
print(price_comparison_counts)

plt.figure(figsize=(8, 6))
sns.barplot(x=price_comparison_counts.index, y=price_comparison_counts.values,
palette='Set2')
plt.title('Price Comparison')
plt.xlabel('Comparison')
plt.ylabel('Count')
plt.show()
```

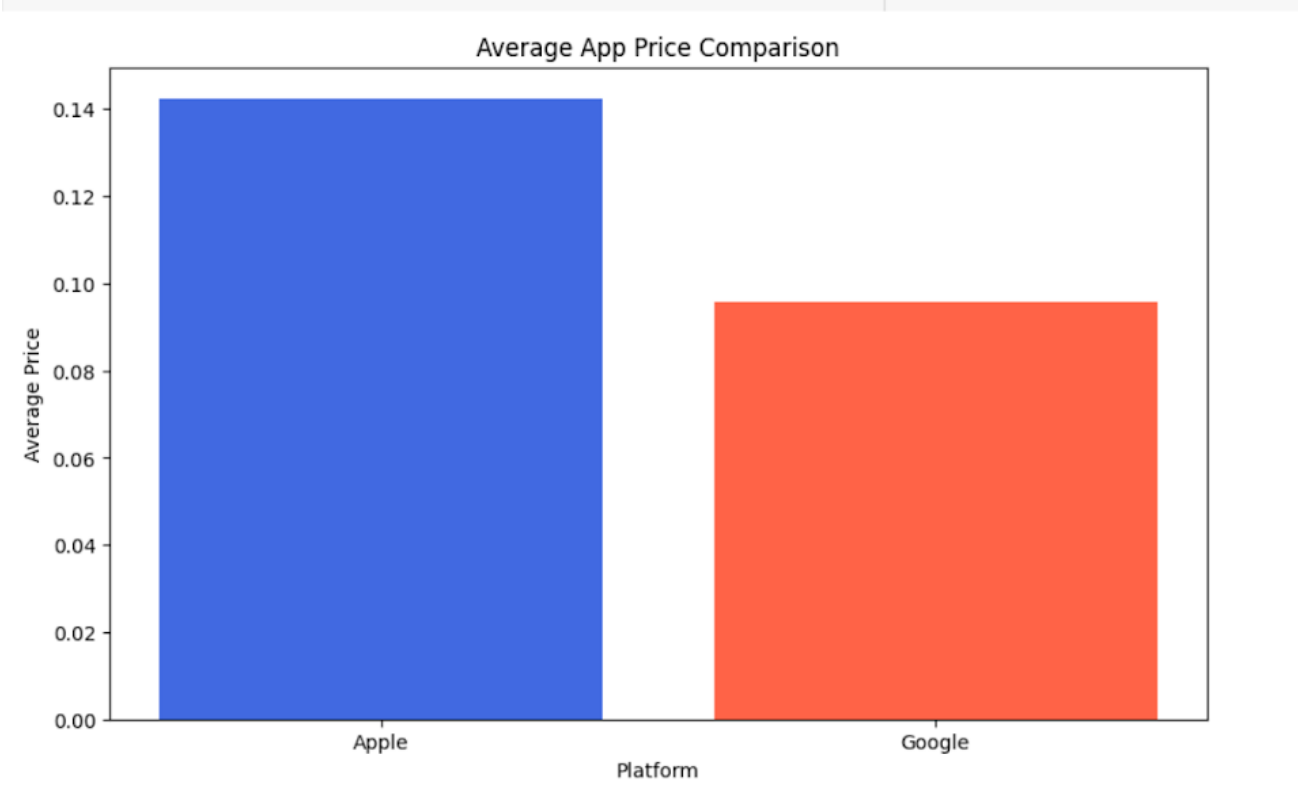
```
Same      356370
Apple       5446
Google      1798
Name: Price_Comparison, dtype: int64
```



```
average_price_apple = data2['Price_x'][data2['Type_x'] == 'Apple'].mean()
average_price_google = data2['Price_y'][data2['Type_y'] == 'Google'].mean()

apple_color = 'royalblue'
google_color = 'tomato'

plt.figure(figsize=(10, 6))
plt.bar(['Apple', 'Google'], [average_price_apple, average_price_google], color=[apple_color,
google_color])
plt.title('Average App Price Comparison')
plt.xlabel('Platform')
plt.ylabel('Average Price')
plt.show()
```



```
target_apps = ['Instagram', 'Facebook','WhatsApp','Snapchat','Telegram','TikTok']

filtered_data = data2[data2['App Name'].isin(target_apps)]

for index, row in filtered_data.iterrows():
    app_name = row['App Name']
    rating_apple = row['Average_User_Rating']
    rating_google = row['Rating']
    price_apple = row['Price_x']
    price_google = row['Price_y']
    size_apple = row['Size_Bytes']
    size_google = row['Size (MB)']

    if rating_apple > rating_google:
        analysis_result = f"'{app_name}' has a higher rating on Apple App Store."
    elif rating_google > rating_apple:
        analysis_result = f"'{app_name}' has a higher rating on Google Play Store."
    else:
        analysis_result = f"'{app_name}' has equal ratings on both platforms."

    if price_apple < price_google:
        analysis_result += f" '{app_name}' has a lower price on Apple App Store."
    elif price_google < price_apple:
        analysis_result += f" '{app_name}' has a lower price on Google Play Store."
    else:
        analysis_result += f" '{app_name}' has equal prices on both platforms."

    if size_apple < size_google:
        analysis_result += f" '{app_name}' has smaller size on Apple App Store.\n"
    elif size_google < size_apple:
        analysis_result += f" '{app_name}' has smaller size on Google Play Store.\n"
    else:
        analysis_result += f" '{app_name}' has equal sizes on both platforms.\n"

    print(analysis_result)
```

'Facebook' has a higher rating on Google Play Store. 'Facebook' has equal prices on both platforms. 'Facebook' has smaller size on Google Play Store.

'Instagram' has a higher rating on Apple App Store. 'Instagram' has equal prices on both platforms. 'Instagram' has smaller size on Google Play Store.

'Snapchat' has a higher rating on Google Play Store. 'Snapchat' has equal prices on both platforms. 'Snapchat' has smaller size on Google Play Store.

'Telegram' has a higher rating on Google Play Store. 'Telegram' has equal prices on both platforms. 'Telegram' has smaller size on Google Play Store.

'TikTok' has a higher rating on Apple App Store. 'TikTok' has equal prices on both platforms. 'TikTok' has smaller size on Google Play Store.

```
data2['Minimum Android'].unique()
```

```
array(['7.0', '4.4', '4.1', '5.0', '2.1', '4.2', '4.0', '3.1', '6.0',
       'Varies with device', '4.3', '5.1', '4.0.3', '2.3.3', '8.0', '3.0',
       '2.3', '2.2', '1.0', '2.0', '7.1', '1.6', '3.2', '2.0.1', '1.5',
       '1.1'], dtype=object)
```

```
data2['Required_IOS_Version'].unique()
```

```
array([13.  ,  9.  ,  7.  ,  8.  , 10.3 ,  9.3 , 11.  , 12.  , 10.  ,
       11.2 , 12.1 , 10.2 ,   nan,  9.2 ,  6.  , 11.1 , 10.11,  8.1 ,
       10.1 , 12.3 ,  7.1 , 11.4 , 14.  , 14.5 ,  9.1 , 12.4 , 13.6 ,
       13.4 , 11.3 , 13.2 , 13.5 , 14.1 ,  8.2 , 12.2 ,  8.4 , 10.8 ,
       14.4 , 13.1 ,  6.1 , 14.3 ,  8.3 ,  4.  , 13.7 , 15.  ,  4.3 ,
       14.2 , 10.15, 10.13,  7.2 , 13.3 , 10.9 , 10.12,  3.  ,  5.1 ,
       10.7 ,  4.2 ,  5.2 , 10.6 ,  5.  , 11.5 , 10.14,  7.9 , 10.4 ,
       12.5 , 10.5 ,  9.4 ,  3.2 ,  7.35,  9.6 ,  5.11, 14.6 ,  2.  ,
        3.1 , 14.7 , 11.24,  4.1 ,  9.5 ,  2.1 ,  2.2 ,  8.5 , 12.16,
        9.9 ,  6.3 ])
```

```
def compare_versions(row):
    android_version = row['Minimum Android']
    ios_version = row['Required_IOS_Version']

    if android_version == 'Varies with device' or pd.isna(ios_version):
        return 'Not Comparable'
    try:
        android_version = float(android_version)
        ios_version = float(ios_version)
    except ValueError:
        return 'Invalid Version Format'

    if android_version <= ios_version:
        return 'Android'
    else:
```

```
    return 'iOS'

data2['Version_Comparison'] = data2.apply(compare_versions, axis=1)
print(data2)
```

```
                        App Name  Size_Bytes  Price_x  Average_User_Rating  \
327808  14F CataloniaÂ´s Elections   21.886719      0.0                    0
77004          3D Meiotic Division  193.672852      0.0                    0
77005          3D Mitotic Division  154.509766      0.0                    3
44254                    A Academy   19.334961      0.0                    0
113759       A Ambassador Limousine   61.852539      0.0                    0
...                             ...         ...      ...                  ...
23134                     myCirqa    6.284180      0.0                    0
1499                      mySAPL   26.045898      0.0                    0
375180                     swaarm   28.590820      0.0                    0
279872           Ä°stanbul Airport  148.147461      0.0                    4
251649              é□□é□æ□ å□□   69.065430      0.0                    0

       Required_IOS_Version     Primary_Genre ContentRating  Free_x  \
327808                 13.0              News       Toddler       1
77004                  9.0         Education       Toddler       1
77005                  7.0         Education       Toddler       1
44254                  8.0         Education       Toddler       1
113759                10.3          Business       Toddler       1
...                    ...               ...          ...     ...
23134                  9.0              Book       Toddler       1
1499                  11.0         Reference       Toddler       1
375180                 9.0         Lifestyle       Toddler       1
279872                11.0            Travel       Toddler       1
251649                 9.0  Health & Fitness       Toddler       1

       Released_Year_x Updated_Year_x  ... Updated_Year_y  Type_y  \
327808            2021           2021  ...           2021  Google
77004             2019           2019  ...           2020  Google
77005             2017           2019  ...           2020  Google
44254             2017           2017  ...           2018  Google
113759            2017           2021  ...           2020  Google
...                ...            ...  ...            ...     ...
23134             2016           2019  ...           2019  Google
1499              2015           2021  ...           2021  Google
375180            2021           2021  ...           2021  Google
279872            2019           2021  ...           2021  Google
251649            2018           2019  ...           2019  Google
```

| | Genre_vs_Category | Diff_Content_Rating | Earlier_Release_Platform | \ |
|---|---|---|---|---|
| 327808 | True | False | Same | |
| 77004 | True | False | Same | |
| 77005 | True | False | Same | |
| 44254 | True | False | Google | |
| 113759 | False | False | Same | |
| ... | ... | ... | ... | |
| 23134 | False | False | Apple | |
| 1499 | True | False | Same | |
| 375180 | True | False | Same | |
| 279872 | True | False | Same | |
| 251649 | False | False | Apple | |

| | Earlier_Updated_Platform | Size_Comparison | Rating_Comparison | \ |
|---|---|---|---|---|
| 327808 | Same | Apple | Google | |
| 77004 | Apple | Apple | Google | |
| 77005 | Apple | Apple | Google | |
| 44254 | Apple | Apple | Same | |
| 113759 | Google | Apple | Same | |
| ... | ... | ... | ... | |
| 23134 | Same | Apple | Google | |
| 1499 | Same | Apple | Google | |
| 375180 | Same | Google | Same | |
| 279872 | Same | Apple | Apple | |
| 251649 | Same | Apple | Same | |

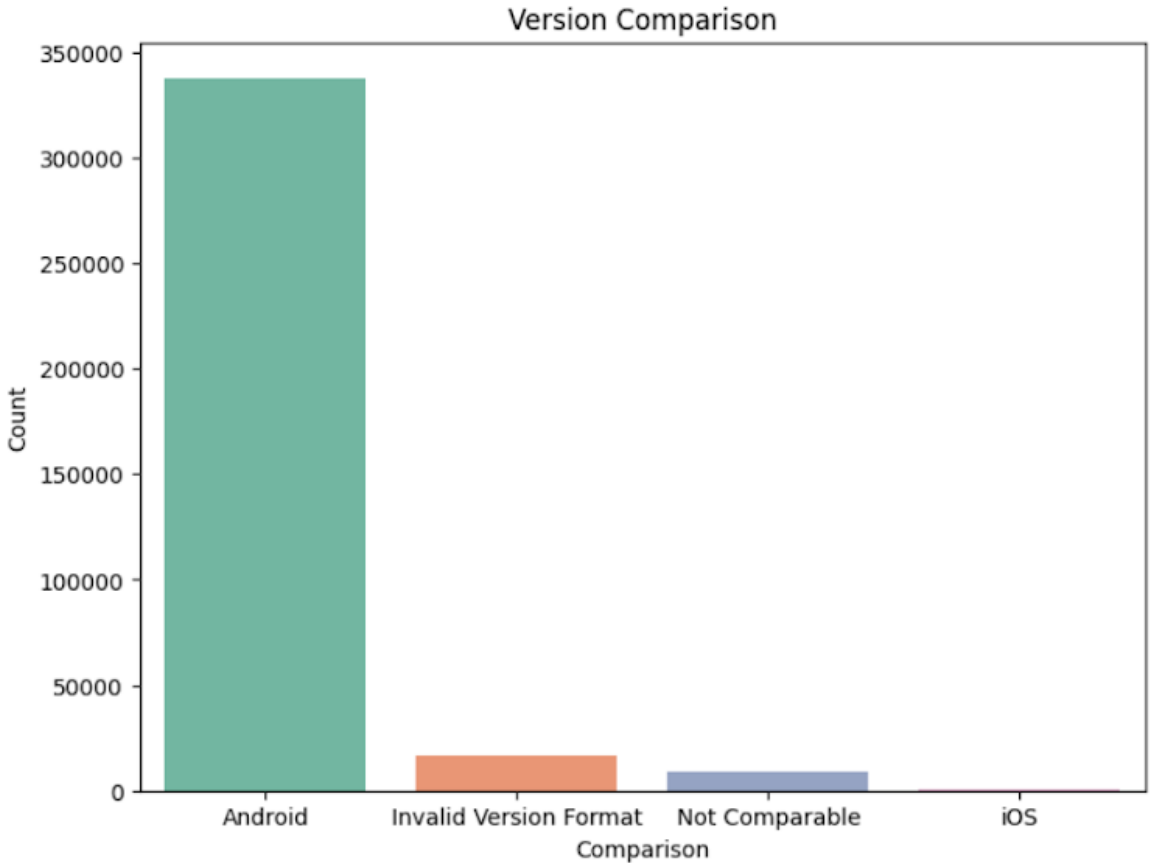
| | Price_Comparison | Version_Comparison |
|---|---|---|
| 327808 | Same | Android |
| 77004 | Same | Android |
| 77005 | Same | Android |
| 44254 | Same | Android |
| 113759 | Same | Android |
| ... | ... | ... |
| 23134 | Same | Android |
| 1499 | Same | Invalid Version Format |
| 375180 | Same | Android |
| 279872 | Same | Android |
| 251649 | Same | Android |

[363614 rows x 29 columns]

```
version_comparison_counts = data2['Version_Comparison'].value_counts()
print(version_comparison_counts)

plt.figure(figsize=(8, 6))
sns.barplot(x=version_comparison_counts.index, y=version_comparison_counts.values,
palette='Set2')
plt.title('Version Comparison')
plt.xlabel('Comparison')
plt.ylabel('Count')
plt.show()
```

```
Android                 337854
Invalid Version Format   16334
Not Comparable            9236
iOS                        190
Name: Version_Comparison, dtype: int64
```



```
def categorize_apps(row):
    if row['Free_x'] and row['Free_y']:
        return 'Both Free'
    elif row['Free_x']:
        return 'Apple Free, Google Paid'
    elif row['Free_y']:
        return 'Apple Paid, Google Free'
    else:
        return 'Both Paid'

data2['Free_Category'] = data2.apply(categorize_apps, axis=1)

print(data2[['App Name', 'Free_x', 'Free_y', 'Free_Category']])
```

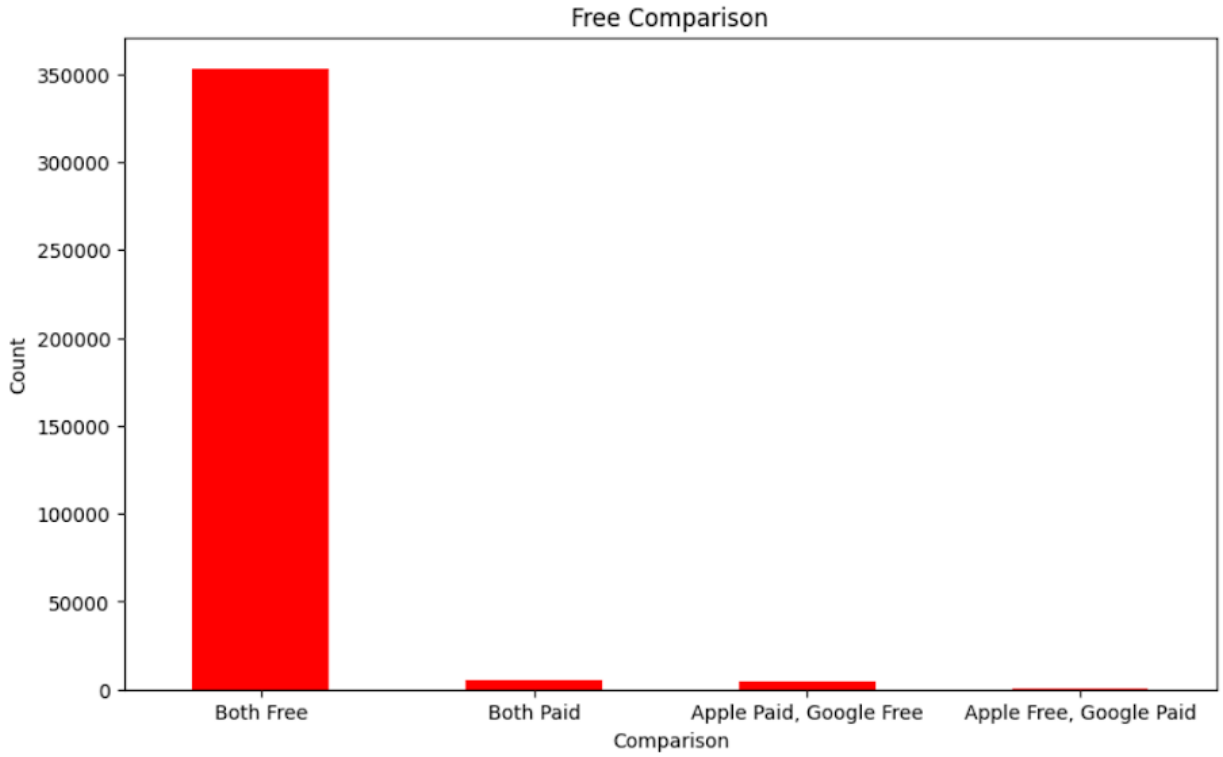
| | App Name | Free_x | Free_y | Free_Category |
|---|---|---|---|---|
| 327808 | 14F CataloniaÂ´s Elections | 1 | 1 | Both Free |
| 77004 | 3D Meiotic Division | 1 | 1 | Both Free |
| 77005 | 3D Mitotic Division | 1 | 1 | Both Free |
| 44254 | A Academy | 1 | 1 | Both Free |
| 113759 | A Ambassador Limousine | 1 | 1 | Both Free |
| ... | ... | ... | ... | ... |
| 23134 | myCirqa | 1 | 1 | Both Free |
| 1499 | mySAPL | 1 | 1 | Both Free |
| 375180 | swaarm | 1 | 1 | Both Free |
| 279872 | Ä°stanbul Airport | 1 | 1 | Both Free |
| 251649 | é□□é□æ□ å□□ | 1 | 1 | Both Free |

[363614 rows x 4 columns]

```
free_comparison_counts = data2['Free_Category'].value_counts()
print(free_comparison_counts)

plt.figure(figsize=(10, 6))
free_comparison_counts.plot(kind='bar', color='red')
plt.title('Free Comparison')
plt.xlabel('Comparison')
plt.ylabel('Count')
plt.xticks(rotation=0)
plt.show()
```

```
Both Free                353201
Both Paid                  5599
Apple Paid, Google Free    4346
Apple Free, Google Paid     468
Name: Free_Category, dtype: int64
```



**COMPARISON**

Based on the comparative analysis of the Google Play Store and Apple App Store data, we can draw the following conclusions:

1. Content Comparison:
   - Genre vs. Category: We compared the genres of apps in both platforms and found that there is a significant overlap in the categories they offer.
   - Content Rating: There is also a high degree of similarity in content ratings between the two platforms.

2. Release and Update Comparisons:
   - Earlier Releases: The majority of apps are released at the same time on both platforms. However, there are cases where Apple App Store gets an app earlier than the other.
   - Earlier Updates: Most apps receive updates at the same time on both platforms. However, there are instances where Google Play Store gets updates earlier.

3. Size and Price Comparisons:
   - Size: Apple's apps tend to have a larger file size compared to Google Play Store apps.
   - Price: App prices are largely similar between the two platforms. Most apps are either free or have the same price on both platforms.

4. User Ratings and Reviews:
   - Average User Ratings: There are differences in average user ratings, with some apps having higher ratings on one platform while others have similar ratings.

5. Specific App Comparisons:
   - We analyzed specific popular apps like Instagram, Facebook, WhatsApp, Snapchat, Telegram, and TikTok. The comparisons vary among these apps, with some having higher ratings, different prices, and varying sizes on different platforms recognizing a general trend being followed by the most popular apps.

6. Operating System Compatibility:
   - We compared the minimum Android version required by apps on the Google Play Store with the required iOS version for Apple apps.
   - The minimum iOS version for apps is higher in comparison to the minimum Android version for the same.
   - Android apps are compatible with lower versions of OS while for iOS apps, a higher version is required for the app to be supported in most cases.

In conclusion, both the Google Play Store and Apple App Store offer a wide range of apps, and the choice between them depends on individual preferences, device compatibility, and specific app features. While there are differences in average user ratings, app sizes, and minor variations in prices, both platforms generally provide similar content. It's important for users to consider their device, content preferences, and app-specific features when choosing between the two platforms.

Based on these findings, it's clear that there are both similarities and differences between Apple and Google app ecosystems. The choice between the two platforms may depend on user preferences for factors like content rating, price, or specific app features.
This analysis can be valuable for developers, marketers, and consumers interested in the app market, as it provides insights into potential areas of improvement, app compatibility, and market competition.
Ultimately, the decision to develop or use an app on one platform over the other should be based on a combination of factors, including target audience, business goals, and the unique features of each platform.